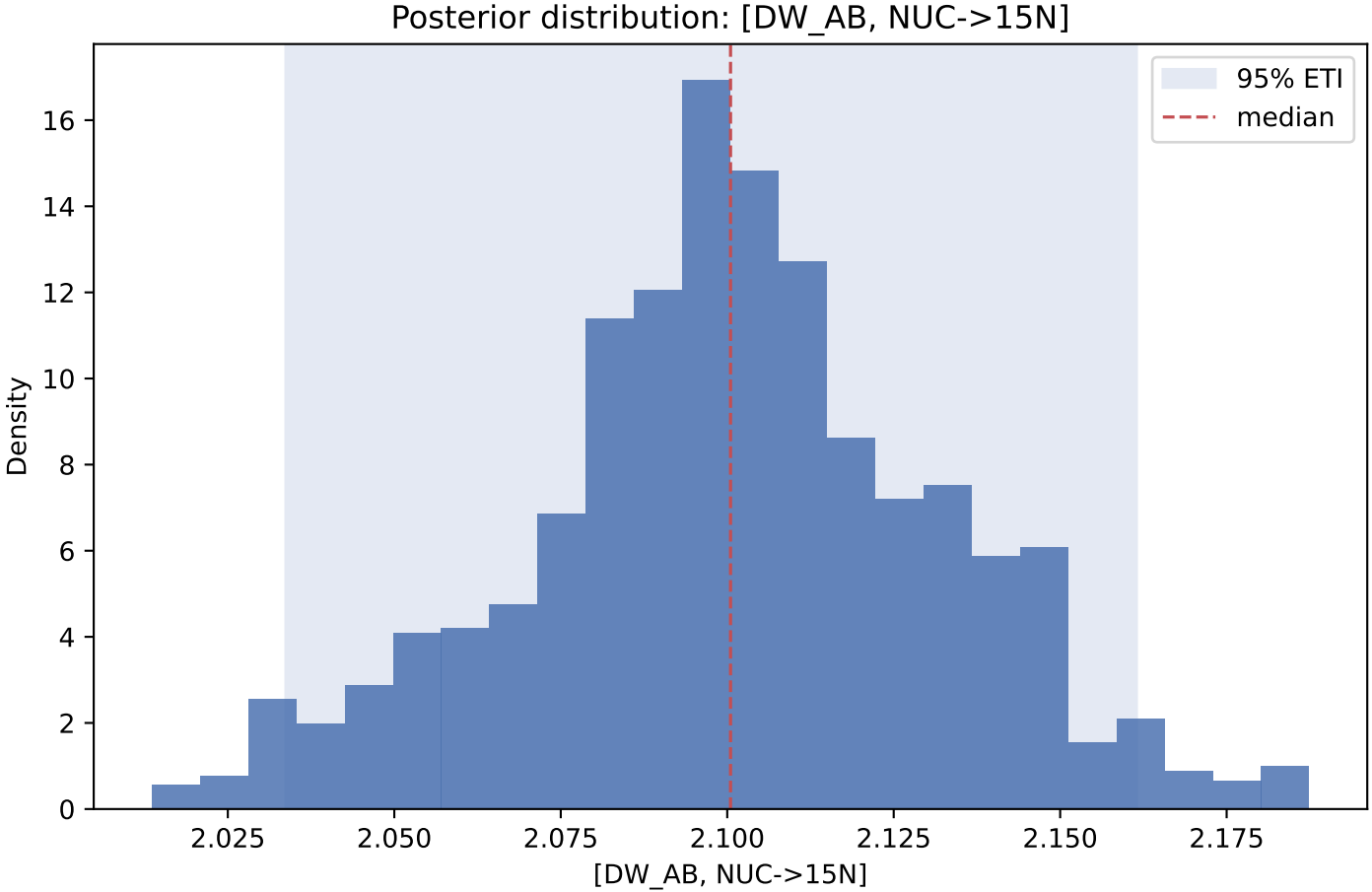
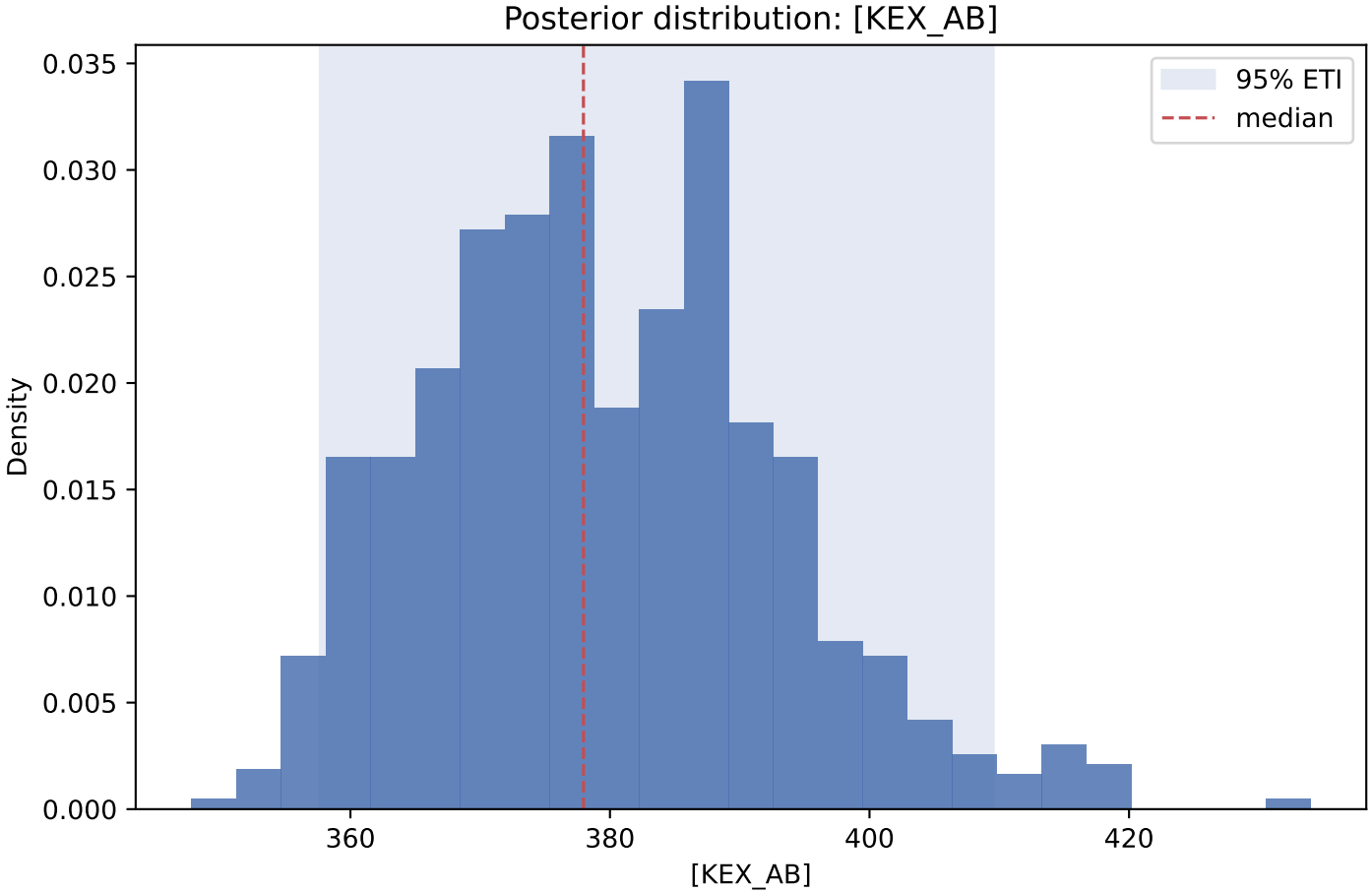
MCMC Diagnostics

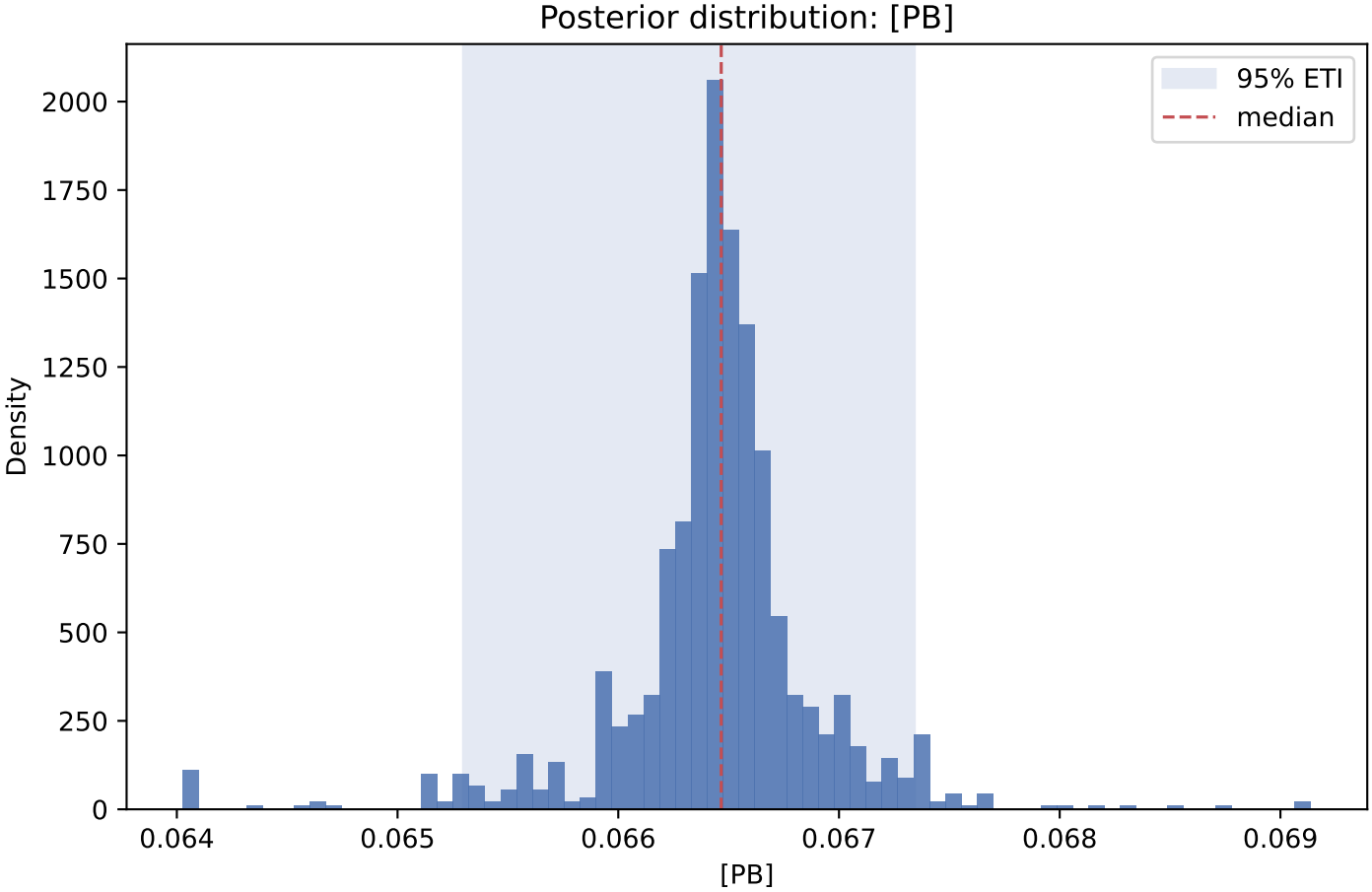
```
Sampler: emcee via ChemEx direct EnsembleSampler
Parameters: 5
Steps: 50
Walkers: 32
Requested burn: "auto"
Discarded steps: 11
Thin: 1
Retained steps: 39
Retained samples: 1248
Acceptance fraction: mean=0.53563, min=0.34, max=0.7
Autocorrelation status: unreliable short chain
Max autocorrelation time: 5.2254
Recommended steps (50 tau): 262
Recommended steps (100 tau): 523
Autocorrelation warning: chain shorter than 50 times the autocorrelation time; tentative estimate rep
Burn-in warning: autocorrelation time estimate is unreliable; tentative automatic burn-in was applied

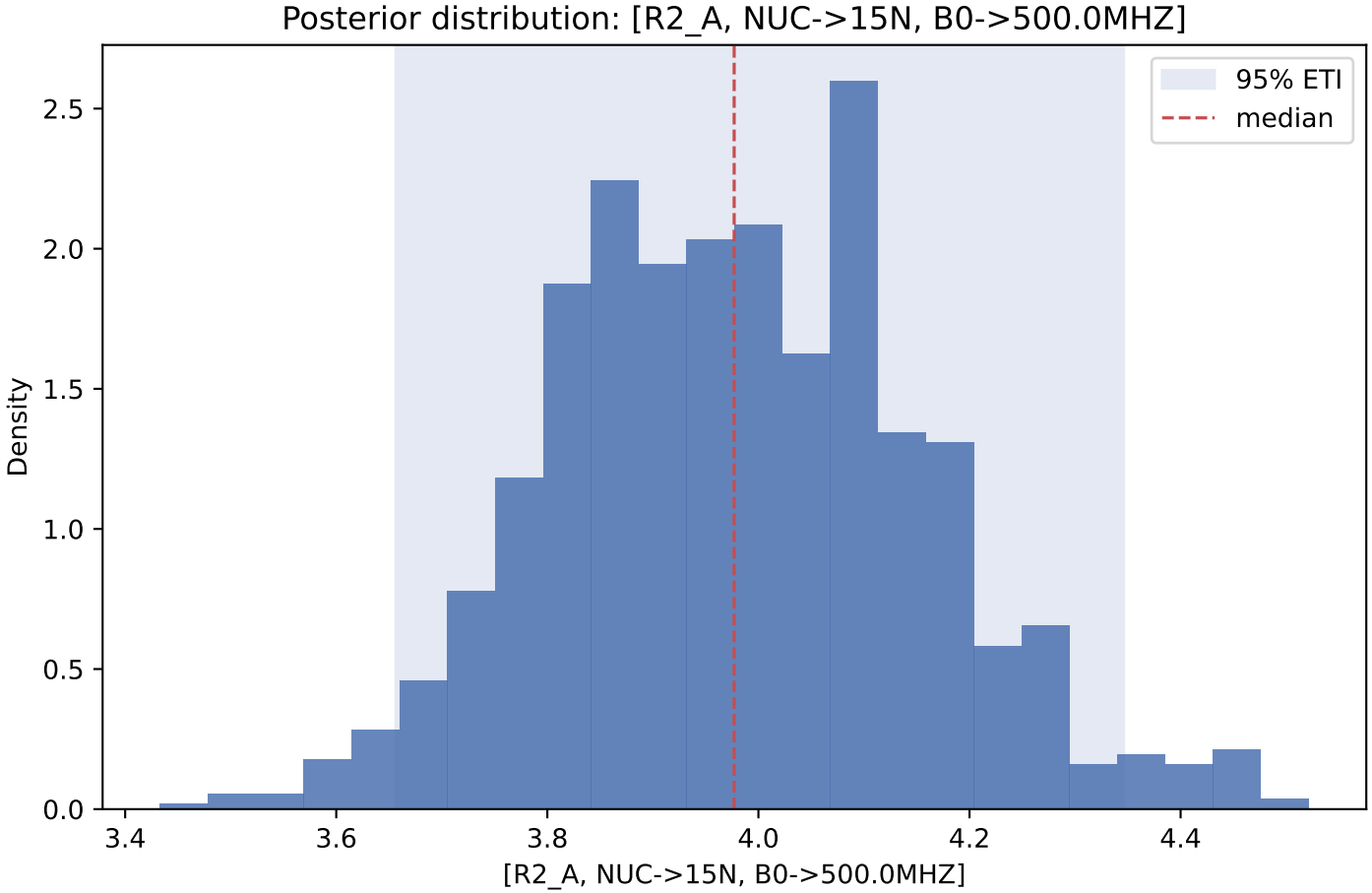
Posterior summaries:
[DW_AB, NUC->15N]: median=2.1005, 95% ETI=[2.0337, 2.1615], ESS=n/a
[KEX_AB]: median=377.96, 95% ETI=[357.66, 409.55], ESS=n/a
[PB]: median=0.066466, 95% ETI=[0.065298, 0.067342], ESS=n/a
[R2_A, NUC->15N, B0->500.0MHZ]: median=3.977, 95% ETI=[3.6564, 4.3461], ESS=n/a
[R2_A, NUC->15N, B0->800.0MHZ]: median=6.3982, 95% ETI=[5.9222, 6.9391], ESS=n/a

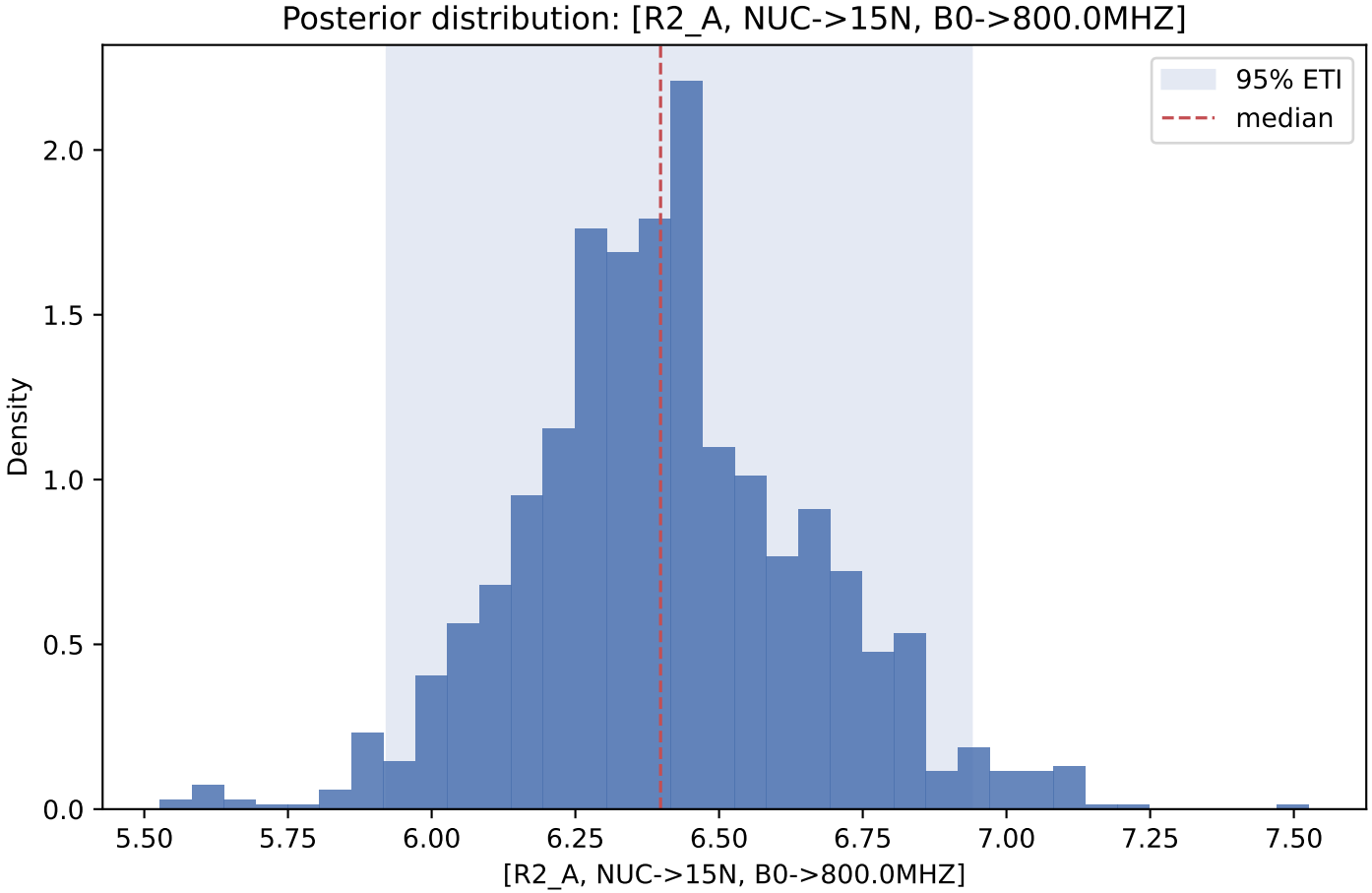
Correlation threshold: |r| >= 0.5
[DW_AB, NUC->15N] vs [KEX_AB]: r=-0.728
```

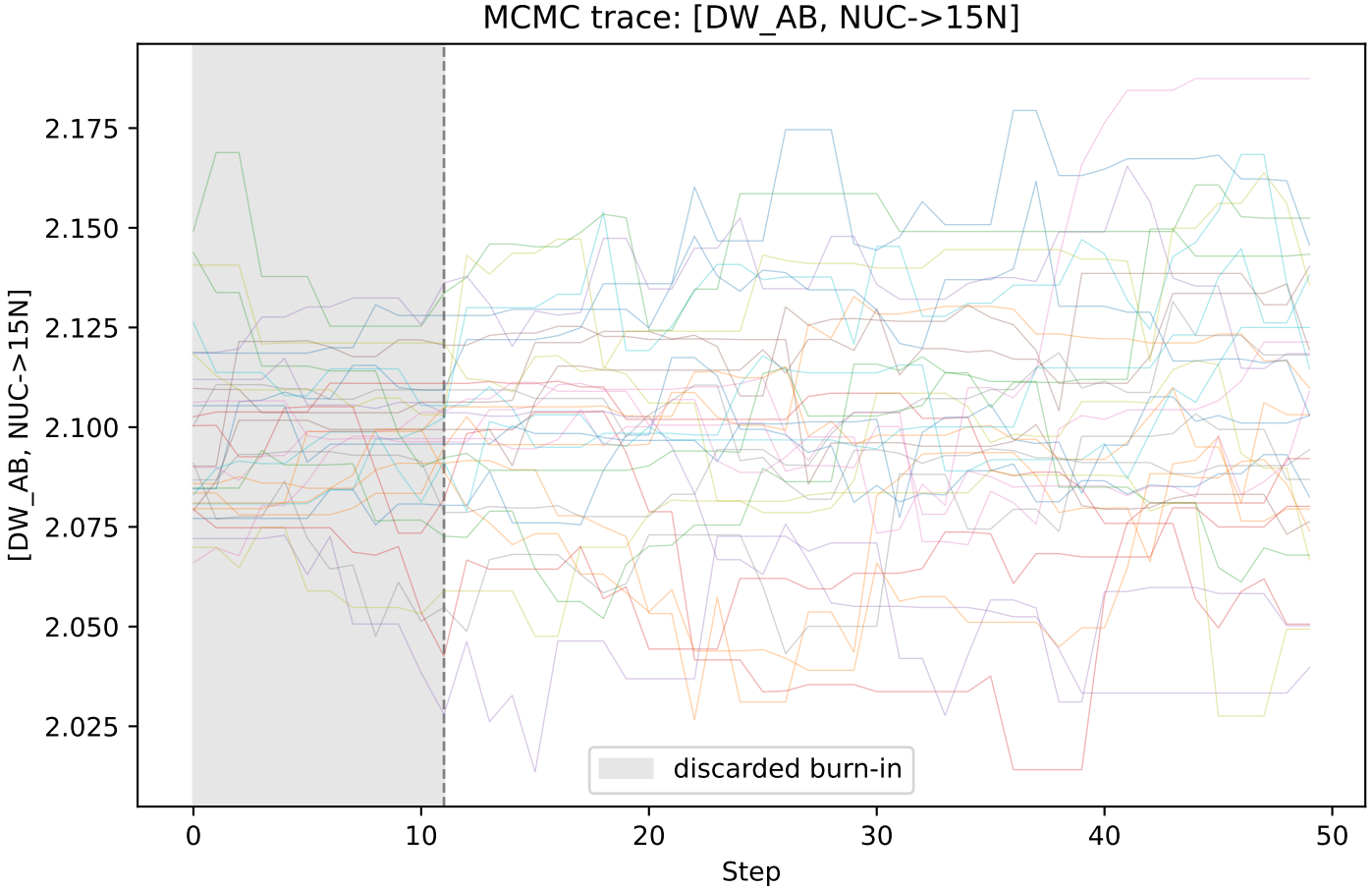

Posterior distribution: [DW_AB, NUC->15N]
95% ETI
median
Density
0
2
4
6
8
10
12
14
16
2.025
2.050
2.075
2.100
2.125
2.150
2.175
[DW_AB, NUC->15N]

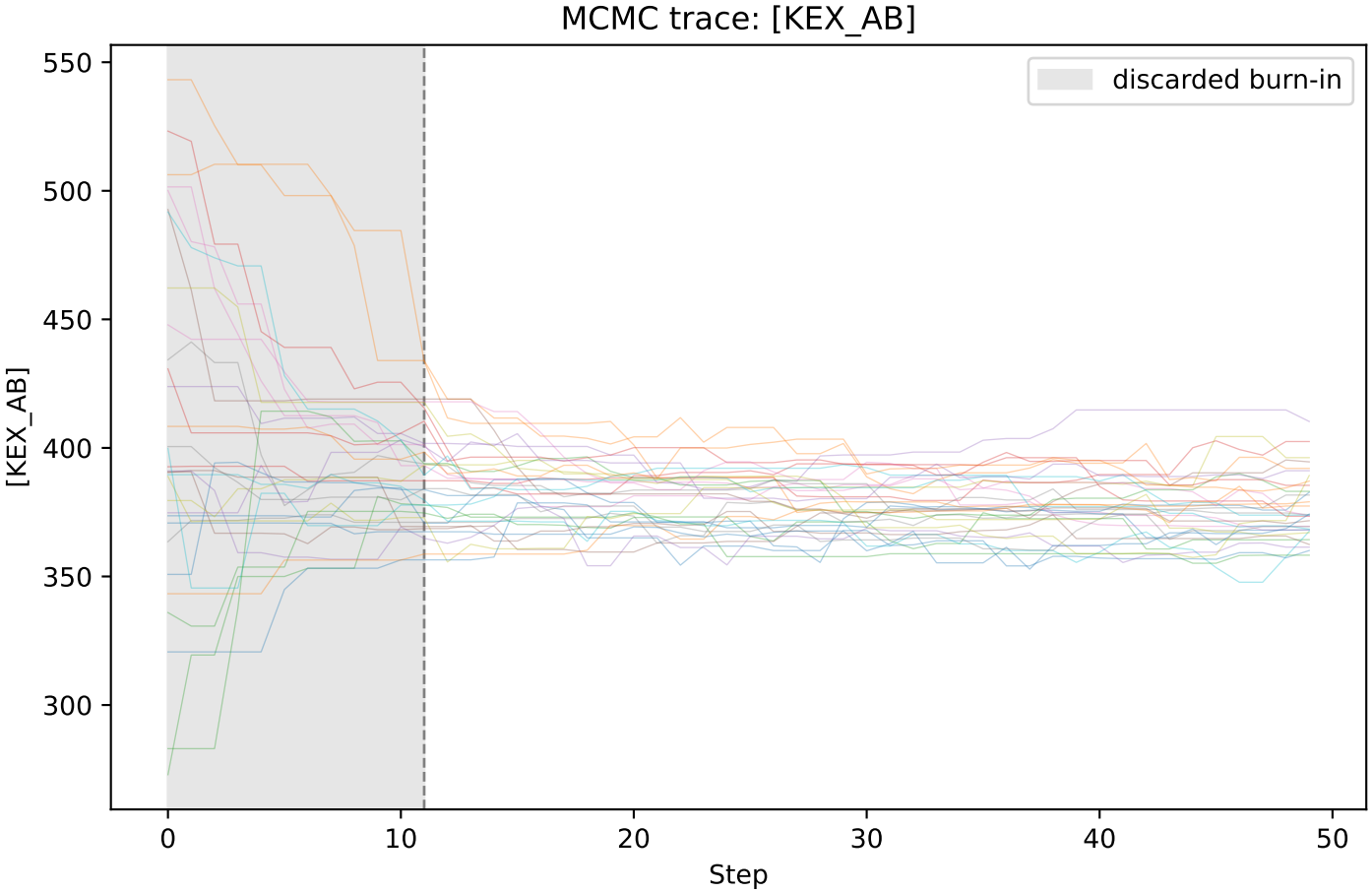

Posterior distribution: [KEX_AB]
95% ETI
median
Density
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
360
380
400
420
[KEX_AB]

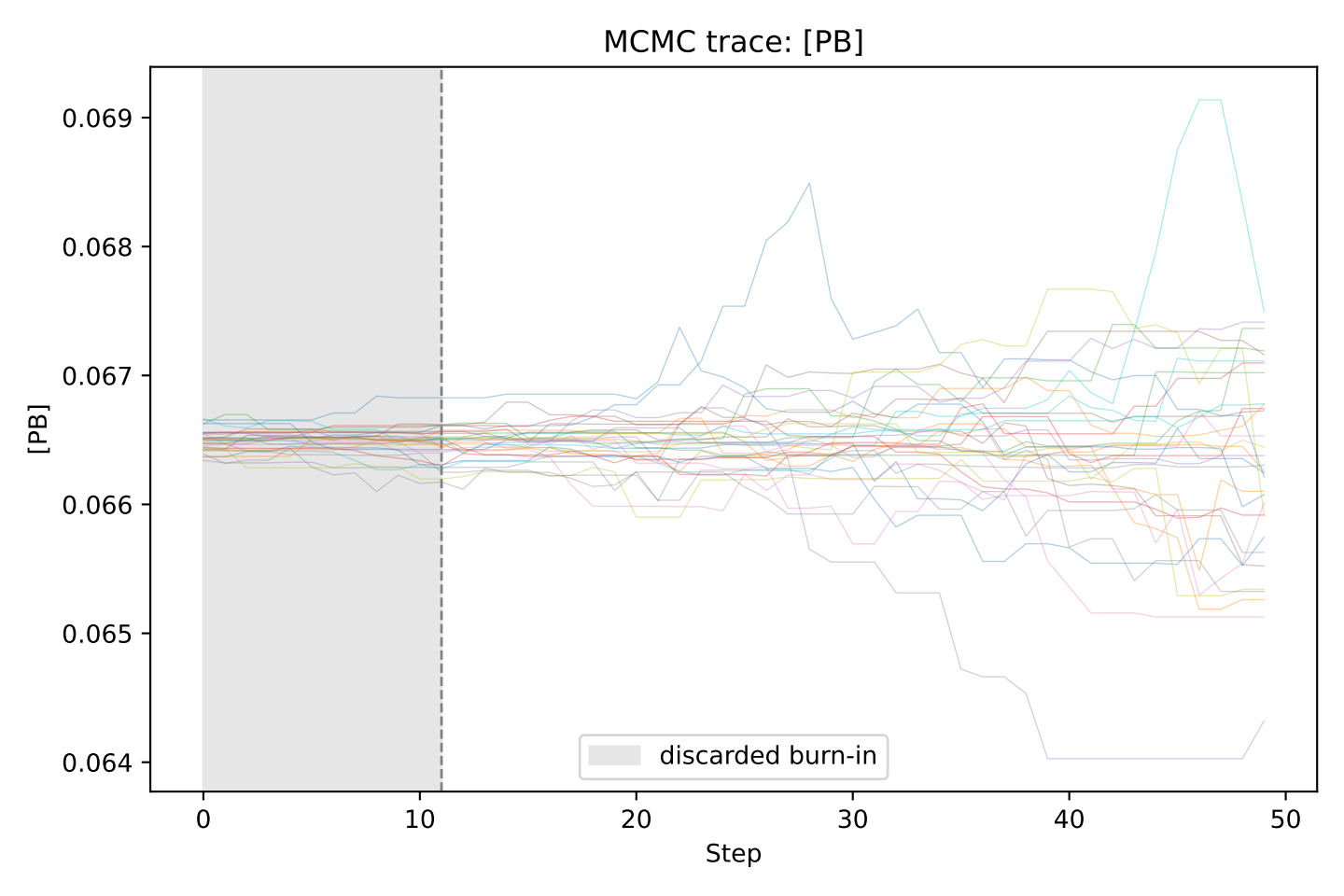

Posterior distribution: [PB]
95% ETI
median
Density
2000
1750
1500
1250
1000
750
500
250
0
0.064
0.065
0.066
0.067
0.068
0.069
[PB]

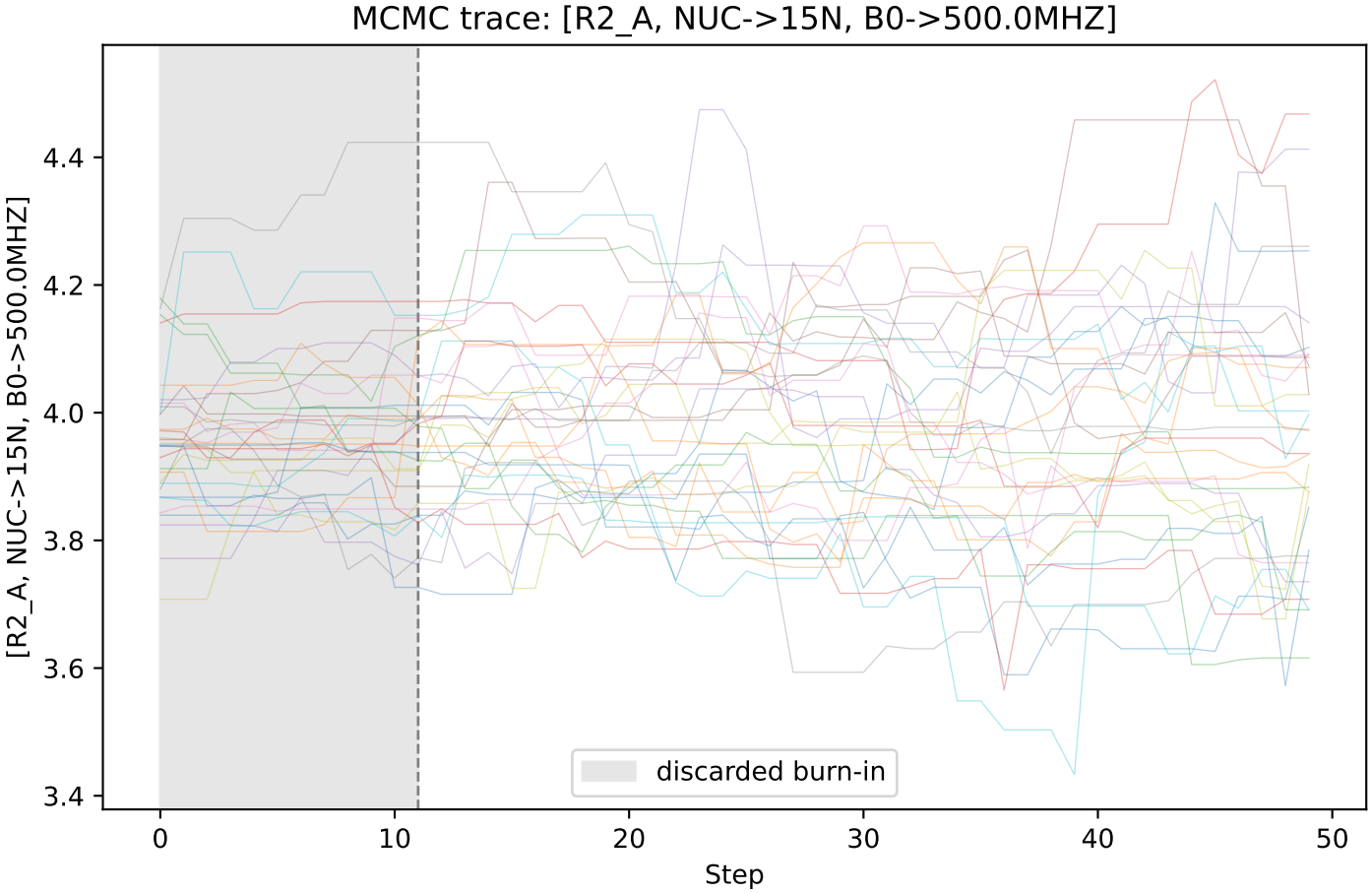

Posterior distribution: [R2_A, NUC->15N, B0->500.0MHZ]
95% ETI
median
Density
0.0
0.5
1.0
1.5
2.0
2.5
3.4
3.6
3.8
4.0
4.2
4.4
[R2_A, NUC->15N, B0->500.0MHZ]

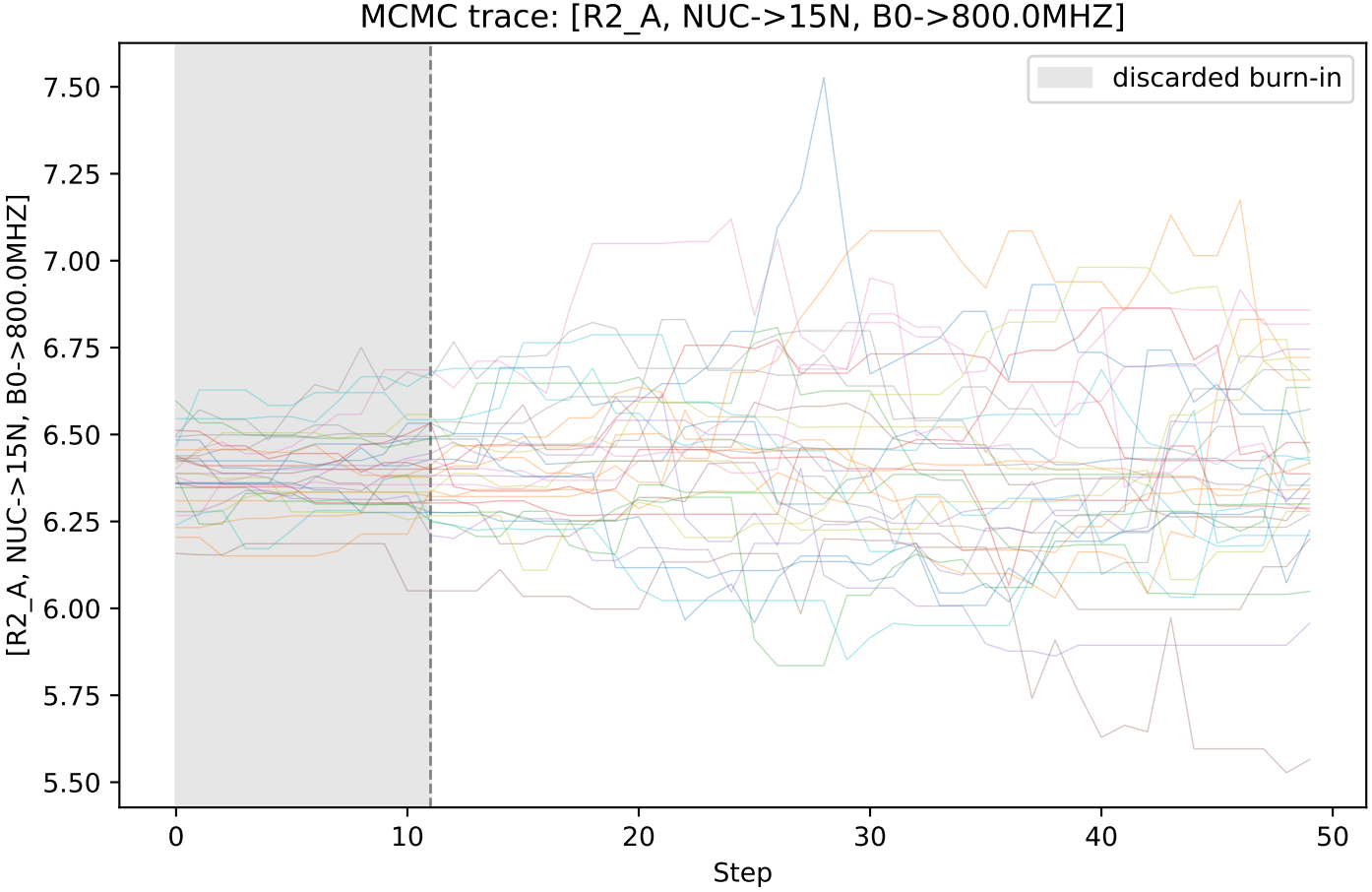

Posterior distribution: [R2_A, NUC->15N, B0->800.0MHZ]
95% ETI
median
Density
0.0
0.5
1.0
1.5
2.0
5.50
5.75
6.00
6.25
6.50
6.75
7.00
7.25
7.50
[R2_A, NUC->15N, B0->800.0MHZ]

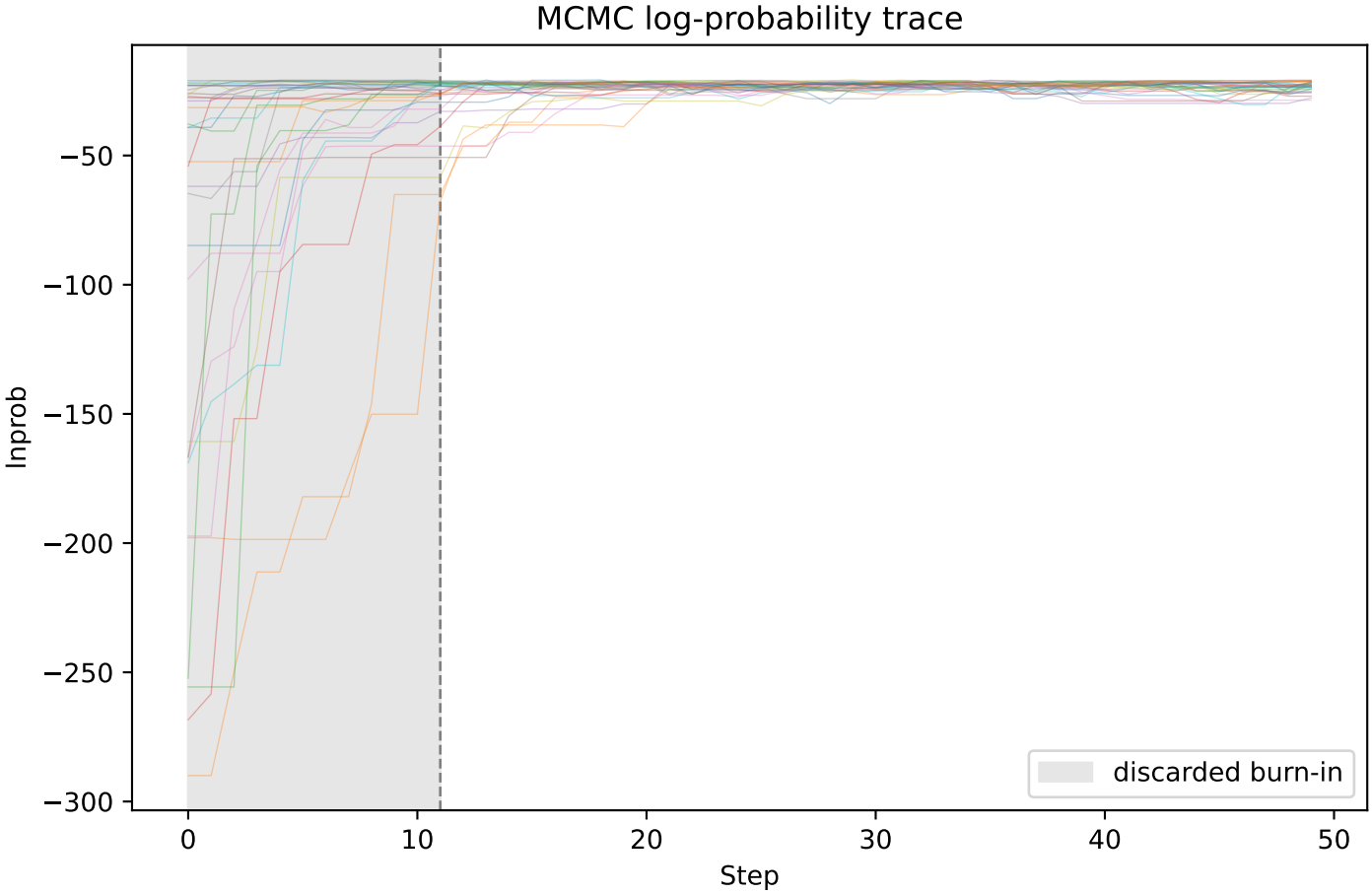

MCMC trace: [DW_AB, NUC->15N]
[DW_AB, NUC->15N]
2.175
2.150
2.125
2.100
2.075
2.050
2.025
0
10
20
30
40
50
Step
discarded burn-in

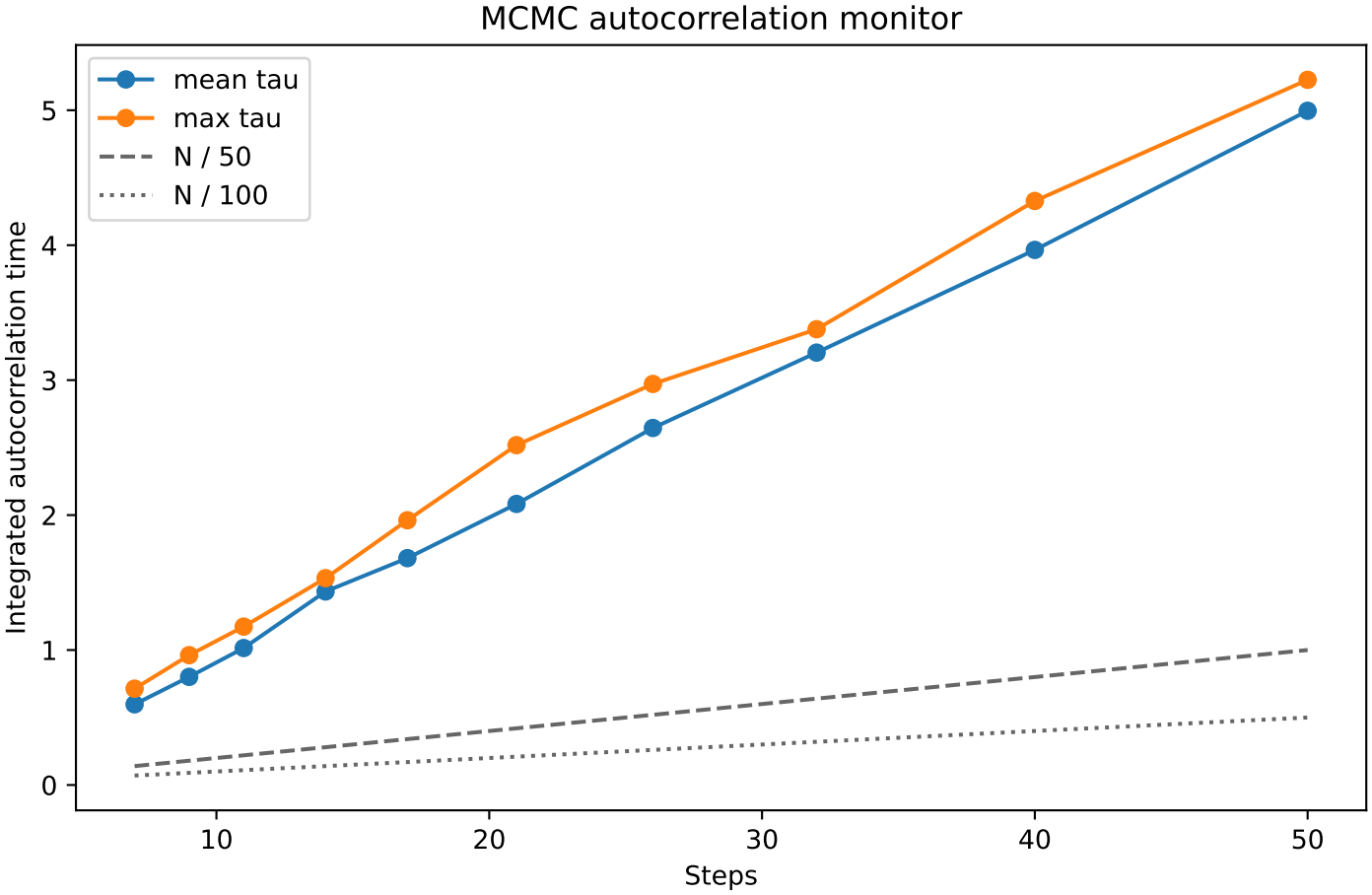

MCMC trace: [KEX_AB]
discarded burn-in
[KEX_AB]
Step
550
500
450
400
350
300
0
10
20
30
40
50

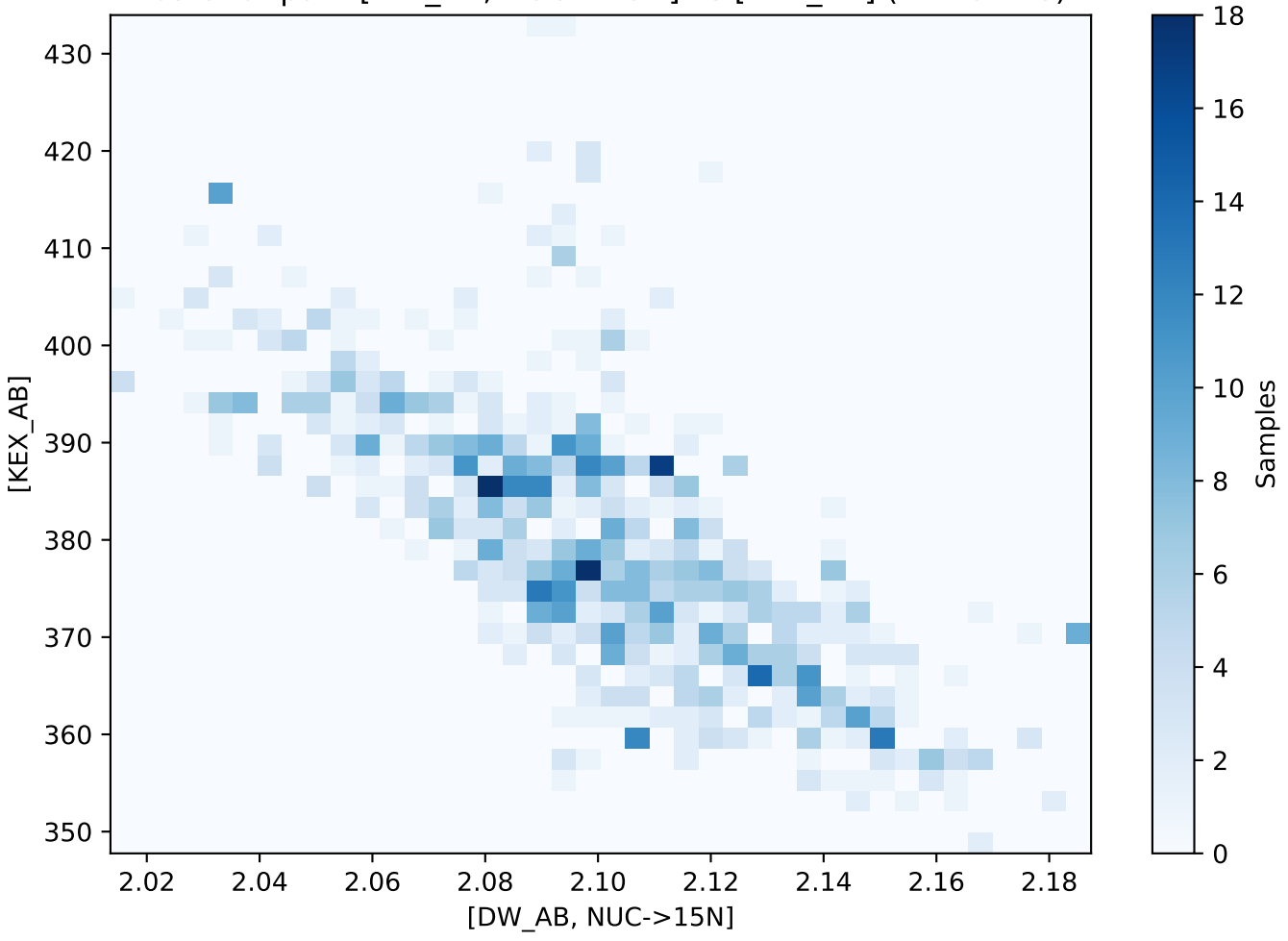

MCMC trace: [PB]
0.069
0.068
0.067
0.066
0.065
0.064
[PB]
0
10
20
30
40
50
Step
discarded burn-in


MCMC trace: [R2_A, NUC->15N, B0->500.0MHZ]
[R2_A, NUC->15N, B0->500.0MHZ]
4.4
4.2
4.0
3.8
3.6
3.4
0
10
20
30
40
50
Step
discarded burn-in


MCMC trace: [R2_A, NUC->15N, B0->800.0MHZ]
discarded burn-in
[R2_A, NUC->15N, B0->800.0MHZ]
7.50
7.25
7.00
6.75
6.50
6.25
6.00
5.75
5.50
0
10
20
30
40
50
Step


MCMC log-probability trace
lnprob
−50
−100
−150
−200
−250
−300
0
10
20
30
40
50
Step
discarded burn-in


MCMC autocorrelation monitor
mean tau
max tau
N / 50
N / 100
Integrated autocorrelation time
Steps
0
1
2
3
4
5
10
20
30
40
50


Posterior pair: [DW_AB, NUC->15N] vs [KEX_AB] (r = -0.728)
[KEX_AB]
430
420
410
400
390
380
370
360
350
2.02
2.04
2.06
2.08
2.10
2.12
2.14
2.16
2.18
[DW_AB, NUC->15N]
Samples
18
16
14
12
10
8
6
4
2
0